Example(s) For Relational Data Model

E1) Suppose each of the following operations is applied directly to the following database state. Discuss all integrity constraints violated by each operation, if any, and the different ways of enforcing these constraints:

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

a) Insert < 'ProductA', 4, 'Bellaire', 2 > into PROJECT.

Violates referential integrity

b) Insert < 'Robert', 'F', 'Scott', '943775543', '21-JUN-42', '2365 Newcastle Rd, Bellaire, TX', M, 58000, '888665555', 1 > into EMPLOYEE.

No constraint violations.

c) Delete the EMPLOYEE tuple with SSN= '987654321'.

Violates referential integrity

d) Delete the PROJECT tuple with PNAME= 'ProductX'.

Violates referential integrity

e) Modify the HOURS attribute of the WORKS_ON tuple with ESSN= '999887777' and PNO= 10 to '5.0'.

No constraint violations

E2) Given the above GARAGE Database state as shown in the below figure. Determine if any constraints will be violated by each of the following operations. If there is a violation then write the type of integrity constraint violation and explain it briefly with the action will be taken.

CUSTOMER

CustomerNo	Fname	Lname	FirstDateSeen
0001	Ahmed	Ali	2/11/2012
0003	Adam	Hassen	15/10/2012
0004	Omar	Elhadi	3/3/2013
0006	Abuallah	Alqhatni	5/4/2013
0007	Ahmed	Abdullah	12/12/2014

VISIT

CarID	VisitDate	VisitTime	VisitReason	VisitSymptoms	MaintenanceCost
C0001	2/8/2015	900	Noise	High Volume	150
C0003	2/8/2015	1200	Engine	Slow start	450
C0006	10/8/2015	1300	Balance	Shift left	6000
C0008	10/8/2015	1500	Air-condition	High temperature	1200

CAR

CarID	CustomerNbr	Model	Year	Mileage	LicensePlate	MechanicalName
C0001	0001	Toyota	2010	112220	XAB246	Ahmed Abdullah
C0003	0003	Nissan	2011	95000	VAV2564	Allan Johns
C0005	0004	Jeep	2010	120000	BYT478	Adam Azizo
C0006	0006	Lexis	2013	75000	NBN254	Ahmed Abdullah
C0008	0007	Ford	2009	135000	TYT235	Allan Cober

a) Insert < 'C0003', ‘0005’, ‘Ford ', ‘2011’, 11524, 'BVB122’,’ Adam Azizo’> into Car.

Violates entity Referential integrity constraint because there is Customer Number 0005 does not exist as an customer .

b) Insert < ‘0005’, 'Jamal ', ‘Berri’, 12/12/2012 > into CUSTOMER .

No constraint violations

c) Delete from CAR the tuple with CarID= 'C0003'.

Violates referential integrity because a tuple exist in the CAR , VISIT relations referencing this tuple.

d) Delete the cusromer whos last name is ALI .

Violates referential integrity because a tuple exist in the Customer and own a car on Car relation.

e) Modify the MaintenanceCost to be 750 for CarID is C0003

No constraint violations.

E3)

Doctor

SSN	FName	LName	Specialty	Years	Phone
556687	Oscar	Wild	ENT	31	050669956
558963	Ahmed	Aljaber	Cardiology	5	055884523
551244	Song	Chen	GP	12	054144236
556394	Jibran	Khalil	Pediatrics	6	055566988
554123	Diana	Moore	Cardiology	26	054223566

Prescription_Medicine

P Id	TradeName	NUnits
P665	CX225	30
P745	Diflao	5
P886	FYY998	21
P589	TAO	6

Patient

SSN	FName	LName	Address	DOB	Doc_SSN
992235	Habib	Khalil	12 32St Jean	15-12-1996	558963
992445	Warda	Jamal	67 Aqid Lotfi	27-6-1968	551244
991178	Patrick	Sail	Olaya 5568	18-1-1998	556687
998841	Mimi	Jalal	223 Brooklyn	29-2-2002	551244
994845	Tong	Mong	884 Kentucky	3-10-1966	554123

Medicine

TradeName	Price	Generic
Hextril	22	F
CX225	13	Y
TAO	85	F
Diflao	98	F
FYY998	26	Y
XSD56F	36	Y

Prescription

Id	Date	Doctor_SSN	Patient_SSN
P665	15-8-2015	554123	994845
P745	17-5-2003	551244	991178
P589	12-12-2012	558963	992235
P886	5-9-1999	554123	994845
P235	2-7-2007	551244	992445
P236	8-11-2009	556394	991178

Using the PRESCRIPTION database in the above page. Determine if any integrity constraints will be violated by each of the following operations. If there is a violation, then write the type of integrity constraint violation and explain it briefly.

1. Insert < 551244, 'Ming', 'Chou', 'Neurology', 19> into Doctor.

Violates entity integrity because SSN 551244 exists already in the relation Doctor

2. Insert < 992827, 'Ali', 'Agha', '34 Manhattan', '23-6-2005', 552039> into Patient.

Violates the referential integrity constraint Mgr_ssn 552039 do not exist as a doctor.

3. Delete the Prescription_Medecine tuples with TradeName= 'TAO'.

No constraint violations.

4. Delete the Medecine tuple with TradeName= 'CX225'.

Violates referential integrity because a tuple in the Prescription_Medecine references this tuple.

5. Modify the TradeName of Medecine tuple 'Diflao' to 'Adol'.

Violates referential integrity because a tuple in the Prescription_Medecine references this tuple.